

Perceived Monetary Policy Rule

Identification and Macroeconomic Implications

Husang Kim • Department of Economics, The University of Texas at Austin • hs.kim@utexas.edu



Takeaways

- ▶ Equilibrium **determinacy** hinges not on whether the central bank satisfies the Taylor principle, **but on whether agents believe it does**.
- ▶ **The perceived rule drives macroeconomic dynamics**, since expectations depend on what agents believe (e.g., the Great Moderation & the post-COVID soft landing).
- ▶ Identifying the perceived rule requires **interest rate nowcasts**, not forecasts.

Simple Model: Why the Perceived Taylor Rule Matters

Setup. Agent knows the structure of the economy but holds a possibly incorrect Taylor rule.

Structure	Rules
$i_t = E_t^P[\pi_{t+1}] + r_t$	$i_t = \phi_p \pi_t$ (perceived)
$r_t = \rho_r r_{t-1} + v_t$	$i_t = \phi_f \pi_t$ (actual)

Perceived dynamics (agent's subjective model, requires $\phi_p > 1$):

$$\pi_{t|t} = \frac{1}{\phi_p - \rho_r} r_t, \quad E_t^P[\pi_{t+1}] = \frac{\rho_r}{\phi_p - \rho_r} r_t$$

- ▶ $\pi_{t|t} = \pi_t$ (FIRE solution) if $\phi_p = \phi_f$.

Actual law of motion (subjective expectations + Fisher equation + actual rule):

$$\pi_t = \frac{1}{\phi_f} \underbrace{E_t^P[\pi_{t+1}]}_{\text{Perception channel}} + \frac{1}{\phi_f} r_t$$

- ▶ Perceived rule governs forward-looking expectations (Perception channel).
- ▶ Actual rule governs pass-through of expectations and shock (Direct policy channel).

Determinacy. A unique bounded equilibrium exists iff

$$\phi_p > 1 \quad (\text{perceived Taylor principle})$$

$\phi_f < 1$ is fully *compatible* with determinacy, provided agents believe $\phi_p > 1$.

Identification: Why Interest Rate Forecasts Introduce Bias

Consider a time-varying Taylor rule:

$$i_t = \phi_{\pi,t} \pi_t + \varepsilon_{M,t}$$

Taking the conditional expectation h periods ahead:

$$\mathbb{E}_t[i_{t+h}] = \mathbb{E}_t[\phi_{\pi,t+h}] \mathbb{E}_t[\pi_{t+h}] + \underbrace{\text{Cov}_t(\phi_{\pi,t+h}, \pi_{t+h})}_{\text{omitted variable}}$$

For a two-state Markov chain $s_t \in \{1, 2\}$, the conditional covariance has the closed form

$$\text{Cov}_t(\phi_{\pi,t+h}, \pi_{t+h}) = P_{j1}^{(h)}(1 - P_{j1}^{(h)})(\phi_{\pi,1} - \phi_{\pi,2})(\pi_{1,t+h}^e - \pi_{2,t+h}^e).$$

When is the covariance non-zero? Agent's joint awareness of:

1. **Possibility of policy regime change.** ($P_{j1}^{(h)} \neq 1$, $\phi_{\pi,1} \neq \phi_{\pi,2}$)
2. **GE effect of monetary policy.** ($\pi_{1,t+h}^e \neq \pi_{2,t+h}^e$)

Sign of the bias. Depends on correlation between $\mathbb{E}_t[\pi]$ and $\text{Cov}_t(\cdot)$.

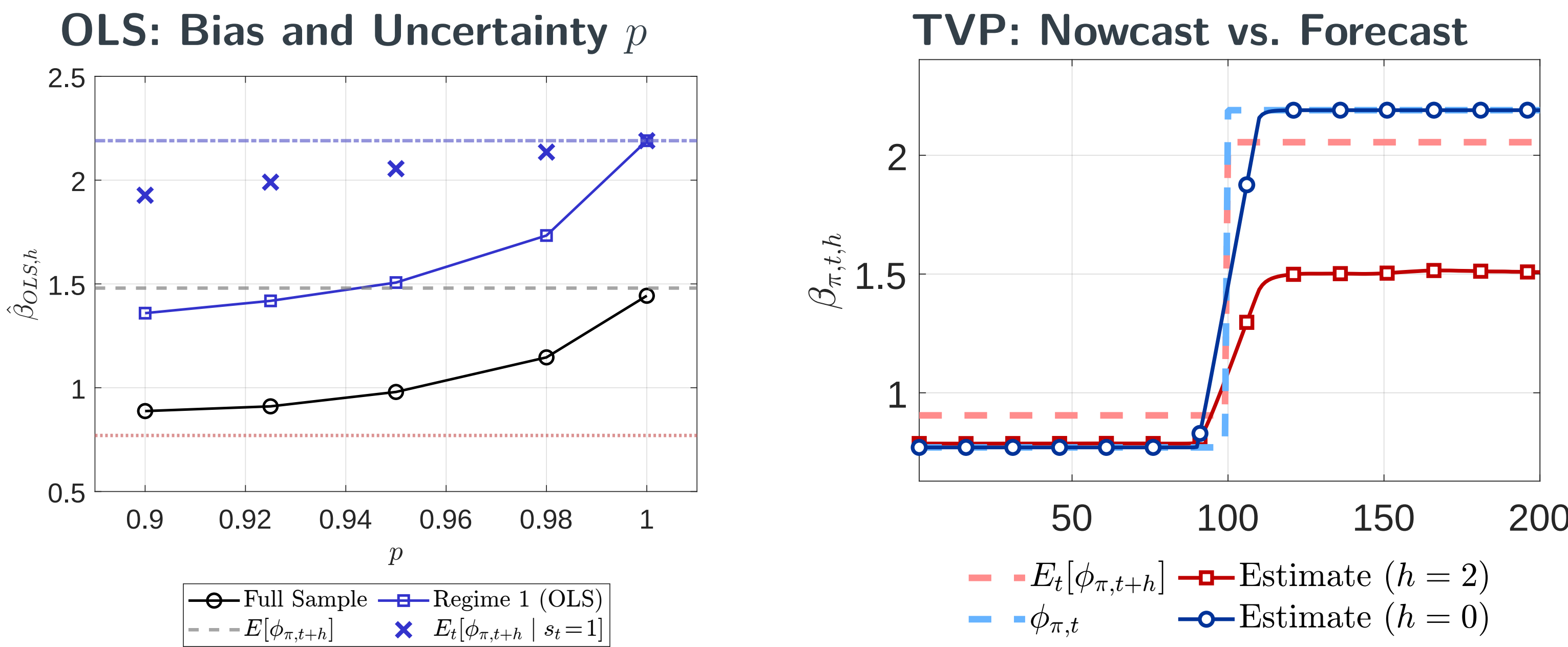
- ▶ In textbook MSNK, when $\mathbb{E}_t[\pi] > 0$, $\text{Cov}_t(\cdot) < 0 \Rightarrow$ **downward bias**.

Resolution. Covariance collapses at $h = 0$. **Apply TVP to interest rate nowcasts.**

Simulation: Nowcast Recovers, Forecast Attenuates

Simulate the model of Lubik-Schorfheide (2004) and Farmer-Waggoner-Zha (2011):

$$\mathbb{E}_t[i_{t+h}] = \beta_h \mathbb{E}_t[\pi_{t+h}] + e_{t,h} \quad (\text{estimated by OLS or TVP})$$

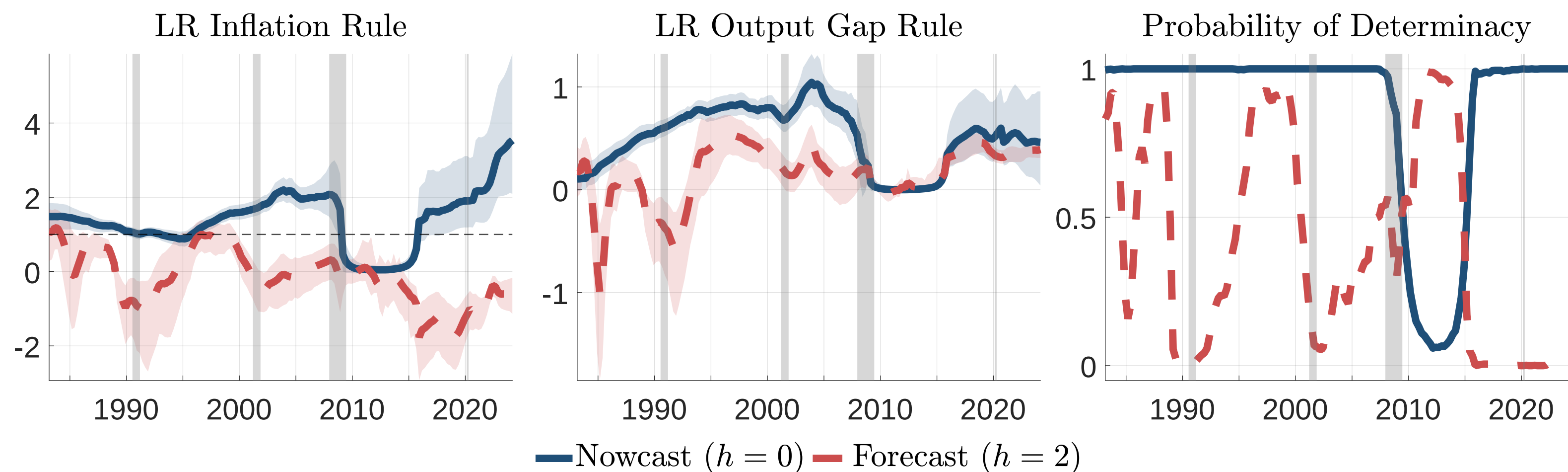


Empirics: Nowcast-Based Estimates Satisfy the Taylor Principle

Data. Blue Chip + SPF survey expectations, 1983–2024 quarterly.

Estimating equation (TVP-SV; Boivin 2005, Coibion-Gorodnichenko 2011):

$$\mathbf{i}_{t+h|t}^{BC} = \beta_{\pi,t,h} \pi_{t+h+1|t}^{BC} + \beta_{x,t,h}(-2 \mathbf{u}_{t+h+1|t}^{BC}) + \beta_{i1,t,h} \mathbf{i}_{t+h-1|t}^{BC} + \beta_{i2,t,h} \mathbf{i}_{t+h-2|t}^{BC} + c_{t,h} + \varepsilon_{t,h}$$



Findings.

- ▶ **Forecast**-based estimates are **attenuated** and **fail the Taylor principle**.
- ▶ **Nowcast**-based estimates are **stable** and **satisfy the Taylor principle**.

SP-MSNK Model: Setup

Perceived model (agent's subjective structural equations):

$$\begin{aligned} x_t &= E_t^P[x_{t+1}] - \tau(i_t - E_t^P[\pi_{t+1}]) + z_{Dt} \\ \pi_t &= \beta E_t^P[\pi_{t+1}] + \kappa x_t + z_{St} \\ i_t &= (1 - \rho_{s_t^p}^P)[\phi_{\pi,s_t^p}^P E_t^P[\pi_{t+1}] + \phi_{x,s_t^p}^P E_t^P[x_{t+1}]] + \rho_{1,s_t^p}^P i_{t-1} + \rho_{2,s_t^p}^P i_{t-2} \end{aligned}$$

Actual Taylor rule (Fed's true reaction function):

$$i_t = (1 - \rho_{s_t^f}^F)[\phi_{\pi,s_t^f}^F E_t^P[\pi_{t+1}] + \phi_{x,s_t^f}^F E_t^P[x_{t+1}]] + \rho_{1,s_t^f}^F i_{t-1} + \rho_{2,s_t^f}^F i_{t-2} + \varepsilon_{Mt}$$

- ▶ The perceived regime $s_t^p \in \mathcal{S}^p$ may differ from the Fed's actual regime $s_t^f \in \mathcal{S}^f$.
- ▶ Subjective transition probabilities defined by **P**.

Perceived law of motion (state-space solution of the perceived model):

$$\mathbf{y}_{t|t} = \bar{\mathbf{A}}_{s_t^p} \mathbf{y}_{t-1} + \bar{\mathbf{B}}_{s_t^p} \varepsilon_t^p$$

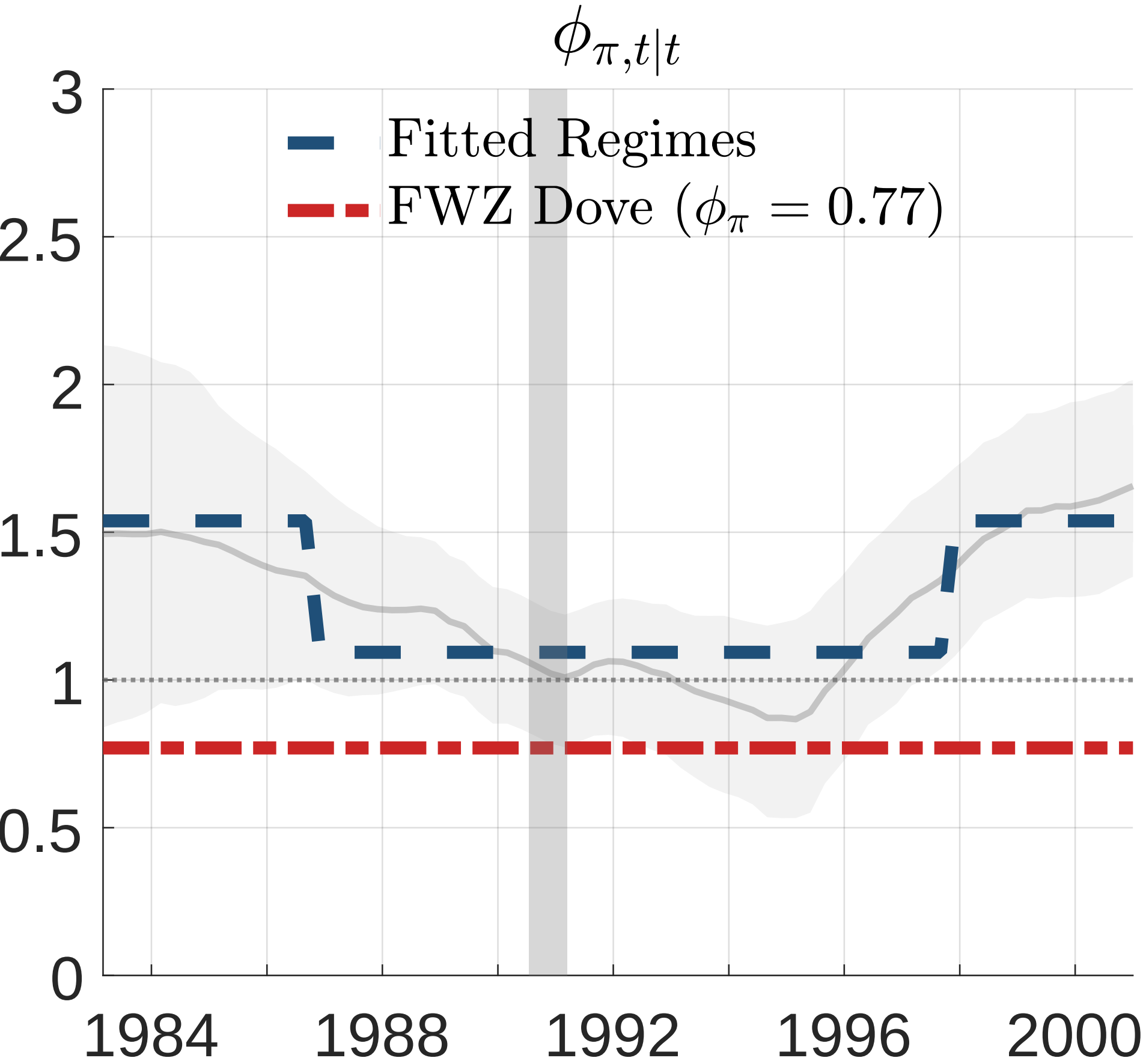
Subjective expectations (under the agent's transition matrix **P**):

$$E_t^P[\mathbf{y}_{t+1}] = \bar{\mathbf{A}}_{s_t^p} \mathbf{y}_{t|t}, \quad \bar{\mathbf{A}}_j \equiv \sum_k \bar{P}_{jk} \bar{\mathbf{A}}_k$$

Actual law of motion (subjective expectations + structural equations + actual rule):

$$\mathbf{y}_t = \mathbf{A}_\ell^{\text{ALM}} \mathbf{y}_{t-1} + \mathbf{B}_\ell^{\text{ALM}} \varepsilon_t, \quad \ell = (s_t^p, s_t^f)$$

Counterfactual: “Good Perceived Policy” Underpinned the Great Moderation



Discretization of perceived rule.

- ▶ A **Hidden Markov Model** is fit to the TVP-estimated $\phi_{\pi,t|t}$ (blue dashed).

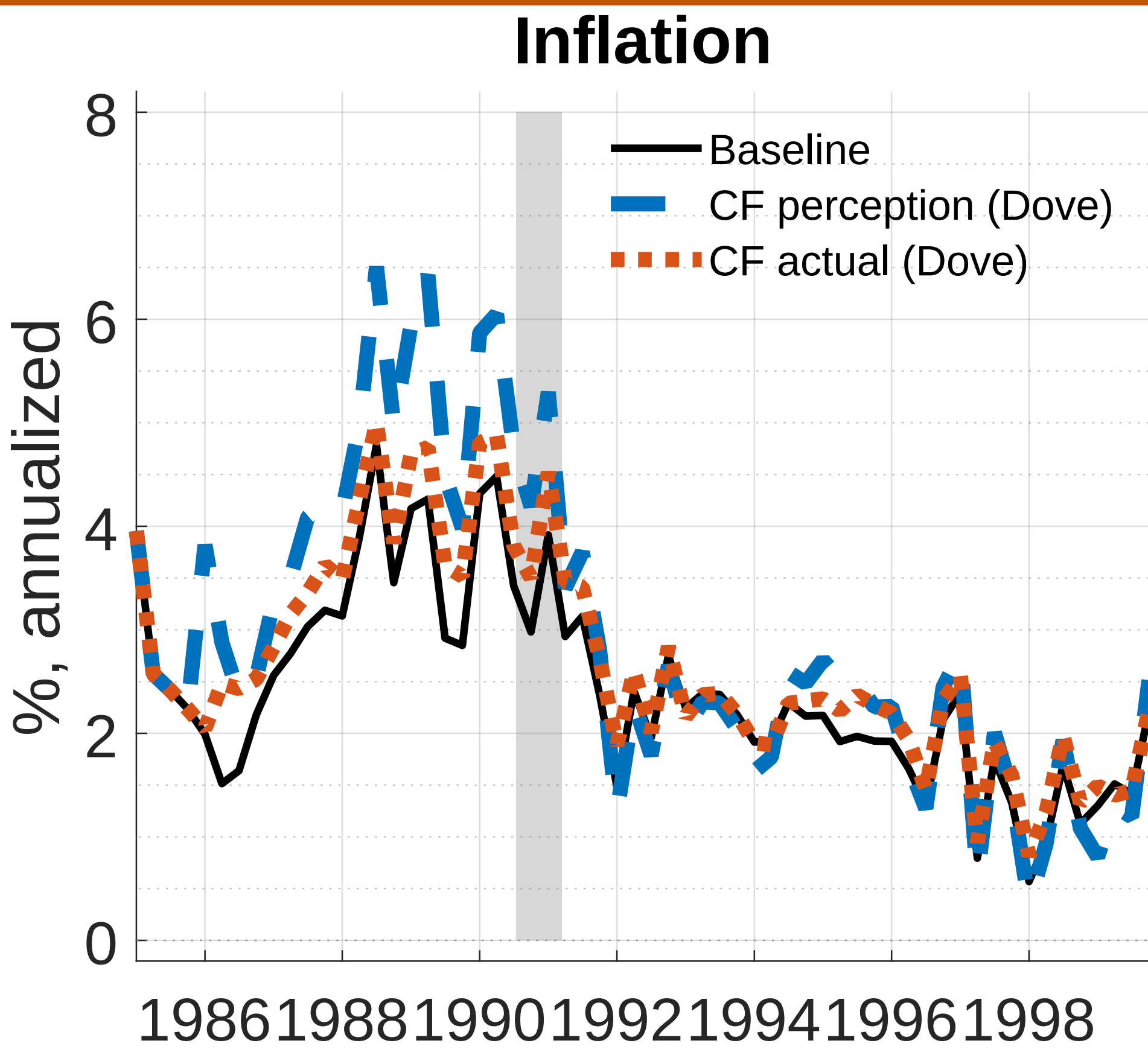
Counterfactual regime.

- ▶ Calibrated from Farmer-Waggoner-Zha (2011): $\phi_p = 0.77$ — a **passive** rule (red dash-dotted).

Counterfactual questions.

- ▶ What if the actual rule stayed at the blue path but the perceived rule deviated to the red path?
- ▶ What if the perceived rule stayed at the blue path but the actual rule deviated to the red path?

How would inflation have evolved in each case?



Two counterfactuals (1986–1999).

- ▶ **Dovish perception** (perceived = Dove, actual unchanged) → **large rise** in inflation volatility.
- ▶ **Dovish actual rule** (actual = Dove, perceived unchanged) → modest rise.

Conclusions.

- ▶ The counterfactual effect through the **perception channel** exceeds that through the direct policy channel.
- ▶ The Great Moderation was sustained by an **anchored hawkish perception**, not by the actual policy stance alone.
- ▶ Application to the **post-COVID inflation surge**: see the paper.
- ▶ **Policy implication.** Effective stabilization requires **managing the public's perception** of the rule.